Mayank Balmoori

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Portfolio • GitHub • LinkedIn

EDUCATION

Institute	Degree / Exam	Year	CGPA/%
Vellore Institute of Technology, Vellore	B.Tech in Computer Science & Engineering	2023 – 2027	8.97
FIITJEE Junior College, Saifabad, Hyderabad	Senior Secondary (Class XII)	2023	94.2%
The Hyderabad Public School, Begumpet, Hyderabad	Secondary (Class X)	2021	97.7%

EXPERIENCE

- **Software Engineer Intern, Providence India** **May 2026 – July 2026**
  - Redesigned file transfer in the Data Exchange Platform (DXP), a healthcare secure-transfer app, with chunk-based streaming at constant memory, removing a ∼2 GB ceiling that blocked 50 GB+ production transfers.
  - Implemented resumable FTPS/SFTP transfers with checkpoint recovery (REST offsets, since FTPS streams are not seekable) and atomic `.tmp` staging with rename, so interrupted jobs resume from the last committed chunk and never leave partial files.
  - Built a T-SQL cron engine (5 stored procedures, 3 scalar functions) with L/W/# syntax and per-job error isolation that scans 2,500+ scheduled jobs against a deployment window in seconds; shipped to production in one week and adopted as the standard pre-release job-impact tool.
  - Added a per-connection SFTP host-key enforcement toggle so server migrations no longer block jobs; wrote design documents as ADRs and shipped through Agile sprints with full code reviews.
- **Full Stack Intern, OriginHash (T-Hub)** **May 2025 – June 2025**
  - Architected and delivered 70% of the Web2 platform in the first month: React frontend and a Node.js/Express backend on MongoDB, with 15+ REST APIs and MongoDB schemas.
  - Built a blockchain-based traceability flow with a document verification system, using Merkle tree hashing for instant verification of agreements.
  - Integrated an OCR model into the Node.js backend for image verification, working directly with the founder.

RESEARCH

- **Research Project (B.Tech. Project-I), VIT Vellore** **July 2026 – Present**
  Working under Dr. Nivitha K (Assistant Professor Sr. Grade 2, VIT Vellore) on a journal manuscript (in preparation) that proposes a new scheduling approach for cloud GPU clusters: it decides which GPU should run each incoming job, balancing energy use against deadline risk. Evaluated through simulation against standard scheduling methods.

PROJECTS

- **Postgres Job Queue** GitHub
  A durable background-job queue on PostgreSQL with Node.js and TypeScript. Workers claim jobs atomically with `FOR UPDATE SKIP LOCKED`, so no two workers take the same job and none wait on each other. A heartbeat lease, fenced updates and a reaper recover jobs from crashed or stalled workers; failed jobs retry with exponential backoff and jitter and end in a dead state that can be requeued; idempotency keys stop duplicate submissions. Verified with 81 integration tests against a real database, including killed workers, a database outage and graceful shutdown, plus a concurrent benchmark with zero duplicate executions. At-least-once delivery by design, and documented as such.
- **Canvix** GitHub
  A real-time collaborative whiteboard with multi-user drawing, shape manipulation, and room-based sync over WebSockets. A dual-canvas design (a preview layer for active drag, resize, and rotate; a committed layer for

finished shapes) avoids full-scene re-renders, and event-based undo/redo records shape operations instead of canvas snapshots to cut memory use. Turborepo monorepo with shared TypeScript packages; open-sourced.

- **QuickBolt** GitHub
  An AI web-app builder (Bolt.new-style): an Express backend uses the Claude API to classify the project type and generate a full project from a prompt; the model's XML output is parsed into file and shell steps, shown in a Monaco file tree, and run in an in-browser WebContainer with live preview and no server-side execution.
- **HD Wallet (ETH + SOL)** GitHub
  A multi-chain wallet that derives Ethereum and Solana accounts from a single BIP39 mnemonic (PBKDF2 seed, BIP32 master key, BIP44 derivation paths) using Ethers.js and Solana Web3.js, built with Next.js and TypeScript; I published a technical write-up of the derivation on Hashnode.
- **EchoTouch** (Microsoft Imagine Cup team project)
  Originated the concept for an AI-powered wearable that translates environmental sounds into haptic feedback patterns for deaf and hard-of-hearing users, and proposed it as our team's Imagine Cup entry.

HACKATHONS

- **Finalist, Caterpillar Hackathon (2026)** *one of fewer than 110 final-round teams*
  In a 10-hour build, a team of two designed and delivered Smart Operator Assistant, an offline-first in-cab assistant for Caterpillar excavators. It plans the operator's day, raises safety alerts with the reason shown, and estimates how long each task will take, with a confidence range. Built for sites with no network: safety alerts are handled on the machine and never wait on a connection.

SKILLS

- **Programming Languages:** TypeScript, JavaScript, Java, C#, C++, C, SQL, T-SQL
- **Frameworks:** React, Next.js, Node.js, Express.js, ASP.NET, .NET, Tailwind CSS
- **AI Tools:** Claude API, LLM integration, prompt engineering, structured output parsing
- **Databases & Tools:** SQL Server, PostgreSQL, MongoDB, Git, GitHub, Docker, AWS, Turborepo
- **APIs & Protocols:** REST APIs, WebSockets, OAuth
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

CERTIFICATIONS

- **Oracle Cloud Infrastructure 2025 Certified Generative AI Professional**
- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate**